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Setup

Load libraries

1. Set `ggplot2` parameters

0.1 A. RT — Words (primary behavioral model)

Question: Does orthographic sensitivity predict efficiency, controlling lexical familiarity?

```
library(lme4)

rt_w <- read.csv("rt_words_frq.csv") %>%
  filter(correct == 1) %>%
  mutate(
    SubjID = factor(SubjID),
    Item   = factor(target),
    logRT  = log(response_time),
    OS     = as.numeric(Dim.2),
    LP     = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )


m_rt_words <- lmer(
  logRT ~ zLogBF + zLogFS + OS + LP +
    zLogBF:OS + zLogFS:OS +
    (1 + zLogBF + zLogFS | SubjID) +
    (1 | Item),
  data = rt_w,
  control = lmerControl(optimizer = "bobyqa")
)

anova(m_rt_words)
```

```
|| Type III Analysis of Variance Table with Satterthwaite's method
||            Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
|| zLogBF    0.40319 0.40319     1 101.849 13.6423 0.0003578 ***
|| zLogFS    0.20034 0.20034     1  93.202  6.7788 0.0107340 *
|| OS        0.10133 0.10133     1  63.877  3.4286 0.0687005 .
|| LP        0.03250 0.03250     1  63.972  1.0995 0.2983154
|| zLogBF:OS 0.02543 0.02543     1 191.849  0.8605 0.3547673
|| zLogFS:OS 0.00575 0.00575     1  78.224  0.1947 0.6602874
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Report: main effects + OS moderation only. LP is included but not emphasized.

0.2 B. RT — Nonwords (primary behavioral model)

Question: Does morphological complexity slow decisions, and is this cost modulated by OS?

```
rt_nw <- read.csv("rt_nwords_frq.csv") %>%
  filter(correct == 1) %>%
  mutate(
    SubjID = factor(SubjID),
    ItemID = factor(ItemID),
    complexity = factor(complexity),  # SIMP / COMP
    logRT  = log(response_time),
    OS     = as.numeric(Dim.2),
    LP     = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

m_rt_nw <- lmer(
  logRT ~ complexity + zLogBF + zLogFS + OS + LP +
    complexity:OS +
    (1 + complexity | SubjID) +
    (1 | ItemID),
  data = rt_nw,
  control = lmerControl(optimizer = "bobyqa")
)

anova(m_rt_nw)
```

```
|| Type III Analysis of Variance Table with Satterthwaite's method
||                Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
|| complexity    1.76964 1.76964     1 61.581 82.5516 5.624e-13 ***
|| zLogBF        0.19870 0.19870     1 98.231  9.2692  0.002991 **
|| zLogFS        0.03278 0.03278     1 96.654  1.5293  0.219217
|| OS            0.10273 0.10273     1 62.351  4.7921  0.032343 *
|| LP            0.00154 0.00154     1 62.908  0.0720  0.789380
|| complexity:OS 0.00300 0.00300     1 59.949  0.1399  0.709712
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

0.3 C. ERP — Words

Run **separately by time window** (this is important).

0.3.1 C1. N250 — Words

Question: Does orthographic sensitivity modulate early sensitivity to morphological family structure?

```
erp_w <- read.csv("erp_dmg_words.csv") %>%
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# ERP words
contrasts(erp_w$family_size) <- contr.sum(2)

m_w_N250 <- lmer(
  value ~ family_size*OS + family_size*LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)

anova(m_w_N250)
```

```
|| Type III Analysis of Variance Table with Satterthwaite's method
||                Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
|| family_size    82.724  82.724     1  1617 30.0635 4.844e-08 ***
|| OS              2.849   2.849     1    57  1.0354   0.31318
|| LP              6.150   6.150     1    57  2.2350   0.14044
|| family_size:OS  7.680   7.680     1  1617  2.7912   0.09498 .
|| family_size:LP  6.409   6.409     1  1617  2.3293   0.12715
```

```
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Follow-up (only if interaction significant):

```r
library(emmeans)
emmeans(m_w_N250, ~ family_size | OS,
        at = list(OS = c(-1, 0, 1)))
```

```
|| OS = -1:
||  family_size emmean    SE   df lower.CL upper.CL
||  Large       -0.900 0.355 58.9   -1.609  -0.1905
||  Small       -1.114 0.355 58.9   -1.824  -0.4051
||
|| OS =  0:
||  family_size emmean    SE   df lower.CL upper.CL
||  Large       -0.585 0.231 58.9   -1.047  -0.1225
||  Small       -0.908 0.231 58.9   -1.370  -0.4461
||
|| OS =  1:
||  family_size emmean    SE   df lower.CL upper.CL
||  Large       -0.269 0.338 58.9   -0.946   0.4080
||  Small       -0.702 0.338 58.9   -1.379  -0.0247
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
```

0.3.2 C2. N400 — Words

Question: Does language proficiency shape semantic integration of family structure?

```r
m_w_N400 <- lmer(
  value ~ family_size*LP + family_size*OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)

anova(m_w_N400)
```

```
|| Type III Analysis of Variance Table with Satterthwaite's method
||                Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
|| family_size    162.349 162.349     1  1617 49.1540 3.464e-12 ***
|| LP               0.000   0.000     1    57  0.0000   0.99577
|| OS              14.982  14.982     1    57  4.5362   0.03752 *
|| family_size:LP   5.944   5.944     1  1617  1.7998   0.17993
|| family_size:OS   5.097   5.097     1  1617  1.5433   0.21430
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Follow-up (core result):

```r
emmeans(m_w_N400, ~ family_size | LP,
        at = list(LP = c(-1, 0, 1)))
```

```
|| LP = -1:
||  family_size emmean    SE   df lower.CL upper.CL
||  Large       0.5499 0.361 58.5  -0.1731    1.273
||  Small       0.0279 0.361 58.5  -0.6951    0.751
||
|| LP =  0:
||  family_size emmean    SE   df lower.CL upper.CL
||  Large       0.5162 0.286 58.5  -0.0556    1.088
||  Small       0.0640 0.286 58.5  -0.5078    0.636
||
|| LP =  1:
||  family_size emmean    SE   df lower.CL upper.CL
||  Large       0.4826 0.375 58.5  -0.2688    1.234
||  Small       0.1001 0.375 58.5  -0.6512    0.852
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
```

```r
emmeans(m_w_N400, ~ family_size | OS,
        at = list(OS = c(-1, 0, 1)))
```

```
|| OS = -1:
||  family_size  emmean    SE   df lower.CL upper.CL
||  Large       -0.2338 0.439 58.5  -1.1121    0.644
||  Small       -0.5968 0.439 58.5  -1.4751    0.282
```

```
||
|| OS =  0:
||  family_size  emmean    SE   df lower.CL upper.CL
||  Large        0.4877 0.286 58.5  -0.0843    1.060
||  Small        0.0358 0.286 58.5  -0.5362    0.608
||
|| OS =  1:
||  family_size  emmean    SE   df lower.CL upper.CL
||  Large        1.2092 0.419 58.5   0.3708    2.048
||  Small        0.6685 0.419 58.5  -0.1699    1.507
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
```

0.4 D. ERP — Nonwords

Again, **separate by time window**.

0.4.1 D1. N250 — Nonwords

Question: Is early morpho-orthographic parsing sensitive to complexity and family size?

```r
erp_nw <- read.csv("erp_dmg_nonwords.csv") %>%
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size),
    complexity = factor(complexity),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# ERP nonwords
contrasts(erp_nw$complexity)  <- contr.sum(2)
contrasts(erp_nw$family_size) <- contr.sum(2)

m_nw_N250 <- lmer(
  value ~ complexity*family_size*OS + complexity*family_size*LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
anova(m_nw_N250)
```

```
|| Type III Analysis of Variance Table with Satterthwaite's method
||                            Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
|| complexity                 87.347  87.347     1  4851 21.4788 3.671e-06 ***
|| family_size                50.567  50.567     1  4851 12.4345 0.0004254 ***
|| OS                          7.456   7.456     1    57  1.8336 0.1810510
|| LP                          6.788   6.788     1    57  1.6692 0.2015774
|| complexity:family_size     10.039  10.039     1  4851  2.4687 0.1162004
|| complexity:OS             198.270 198.270     1  4851 48.7553 3.291e-12 ***
|| family_size:OS              5.477   5.477     1  4851  1.3468 0.2458962
|| complexity:LP             312.301 312.301     1  4851 76.7958 < 2.2e-16 ***
|| family_size:LP              4.184   4.184     1  4851  1.0290 0.3104513
|| complexity:family_size:OS  21.828  21.828     1  4851  5.3676 0.0205553 *
|| complexity:family_size:LP   5.888   5.888     1  4851  1.4478 0.2289440
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

0.4.2 D2. N400 — Nonwords (your strongest ERP result)

Question: Does language proficiency determine how familiarity modulates complexity costs?

```r
m_nw_N400 <- lmer(
  value ~ complexity*family_size*LP + complexity*family_size*OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)

anova(m_nw_N400)
```

```
|| Type III Analysis of Variance Table with Satterthwaite's method
```

```
||                          Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
|| complexity               355.67  355.67     1  4851 58.3080 2.683e-14 ***
|| family_size                11.59   11.59     1  4851  1.8997 0.1681781
|| LP                          0.01    0.01     1    57  0.0020 0.9644627
|| OS                          7.23    7.23     1    57  1.1851 0.2809041
|| complexity:family_size     80.25   80.25     1  4851 13.1555 0.0002896 ***
|| complexity:LP             157.63  157.63     1  4851 25.8424 3.844e-07 ***
|| family_size:LP             36.46   36.46     1  4851  5.9771 0.0145283 *
|| complexity:OS              74.31   74.31     1  4851 12.1824 0.0004867 ***
|| family_size:OS              5.56    5.56     1  4851  0.9121 0.3396090
|| complexity:family_size:LP  32.11   32.11     1  4851  5.2635 0.0218195 *
|| complexity:family_size:OS   6.56    6.56     1  4851  1.0761 0.2996187
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Critical follow-up:

```r
emmeans(
  m_nw_N400,
  ~ complexity*family_size | LP,
  at = list(LP = c(-1, 0, 1))
)
```

```
|| LP = -1:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large        0.0596 0.401 Inf    -0.727    0.8458
||  Simple     Large       -0.9885 0.401 Inf    -1.775   -0.2023
||  Complex    Small       -0.2403 0.401 Inf    -1.027    0.5459
||  Simple     Small       -0.6191 0.401 Inf    -1.405    0.1671
||
|| LP =  0:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large       -0.0523 0.317 Inf    -0.674    0.5696
||  Simple     Large       -0.7316 0.317 Inf    -1.353   -0.1098
||  Complex    Small       -0.3597 0.317 Inf    -0.982    0.2621
||  Simple     Small       -0.5991 0.317 Inf    -1.221    0.0227
||
|| LP =  1:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large       -0.1642 0.417 Inf    -0.981    0.6529
||  Simple     Large       -0.4748 0.417 Inf    -1.292    0.3423
||  Complex    Small       -0.4791 0.417 Inf    -1.296    0.3379
||  Simple     Small       -0.5791 0.417 Inf    -1.396    0.2379
||
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95
```

```r
emmeans(
  m_nw_N400,
  ~ complexity*family_size | OS,
  at = list(OS = c(-1, 0, 1))
)
```

```
|| OS = -1:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large       -0.3161 0.487 Inf    -1.271   0.63900
||  Simple     Large       -1.3376 0.487 Inf    -2.293  -0.38248
||  Complex    Small       -0.6291 0.487 Inf    -1.584   0.32600
||  Simple     Small       -1.0509 0.487 Inf    -2.006  -0.09582
||
|| OS =  0:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large       -0.0577 0.317 Inf    -0.680   0.56437
||  Simple     Large       -0.7679 0.317 Inf    -1.390  -0.14581
||  Complex    Small       -0.3650 0.317 Inf    -0.987   0.25703
||  Simple     Small       -0.6182 0.317 Inf    -1.240   0.00386
||
|| OS =  1:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large        0.2007 0.465 Inf    -0.711   1.11240
||  Simple     Large       -0.1981 0.465 Inf    -1.110   0.71352
||  Complex    Small       -0.1009 0.465 Inf    -1.013   0.81073
||  Simple     Small       -0.1855 0.465 Inf    -1.097   0.72621
||
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95
```